Al Muqshith Shifan

Applied ML / RL Developer • Systems Engineering • Research Assistant

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Professional Summary

MSc Computer Science candidate specializing in **software design for artificial intelligence and reinforcement learning (RL)** with hands-on experience building DRL training infrastructure (multi-head PPO critics, custom PyTorch environments, persona-based reward systems) and programmable-network research systems on **Intel Tofino P4** silicon. CCNA-certified; comfortable in low-level Linux/Unix stacks. Looking to contribute to GPU-native ML tooling.

Education

Ontario Tech University — Oshawa, ON
MSc in Computer Science – Software Design (AI/RL Focus) — *May 2026 – Present*

Ontario Tech University — Oshawa, ON
Bachelor of IT – Networking & Cybersecurity; GPA 3.8/4.2 (Dean's List 2026) — *Graduated Jun 2026*

Work Experience

Teaching Assistant – Networking I & Advanced Networking II — Sep 2025 – Present
Ontario Tech University — *Oshawa, ON*
- Led weekly lab sessions for ∼120 undergraduates across two networking courses, guiding hands-on Cisco IOS configuration of routers, switches, and stateful firewalls; mentored students through TCP/IP, VLSM subnetting, and routing protocol debugging in office hours.

Network Administrator – Summer Research Assistant — Jun 2025 – Sep 2025
Ontario Tech University — *Oshawa, ON*
- Engineered programmable-network research infrastructure on **Intel Tofino silicon P4 switches**, supporting faculty research on in-network telemetry and traffic analysis.
- Authored custom P4 data-plane programs extracting per-flow header and timing telemetry at line rate, delivering the lab's first working Tofino-deployed implementation.
- Diagnosed and patched recurring Linux kernel and driver issues across P4 control-plane servers, reducing reported system crashes by ∼30% over a 3-month window.

Cyber Security Analyst — Jun 2025 – Sep 2025
YYC Beeswax Ltd — *Remote*
- Performed vulnerability assessments on a WooCommerce + Stripe e-commerce stack, identifying 15 exploitable issues across customer-facing endpoints with prioritized remediation steps.
- Audited 30+ WordPress/WooCommerce plugins against CVE databases and authored a hardening checklist adopted as the engineering team's pre-deploy review gate.

Networking Lab Technician — Sep 2024 – Jun 2025
Ontario Tech University — *Oshawa, ON*
- Cut lab network downtime by ∼20% for 50+ daily users by isolating switching bottlenecks and reconfiguring VLAN trunking; maintained 20+ workstations and DHCP/DNS/NTP services, resolving Tier-2 tickets.

Projects

RIDGE – Reactive Inter-persona Dynamic Goal Engine | *Python, PyTorch, Crafter, PPO* — 2026
- Second author on a five-author deep RL architecture for state-conditioned reward blending across four persona objectives (Explorer, Survivor, Craftsman, Warrior) on Crafter. **Accepted at IEEE Conference on Games (CoG) 2026**, Madrid, Spain. GitHub: Code-SorceryLab/RIDGE.
- Designed a **multi-head PPO critic** (shared CNN encoder, four persona-specific value heads) trained jointly with smooth sigmoid weight blending over a 6-dim game-state vector to address PPO non-stationarity under dynamic reward weighting.
- Built the full training pipeline in PyTorch with TensorBoard logging and YAML configs; ran the 1M-step Crafter budget across four fixed-persona baselines and a five-point sharpness ablation, concentrating seeds ($n = 3$) on the central conditions.
- Result: a single agent acquires both branches of Crafter's wood-tier crafting/combat split (0.45 pickaxe, 0.51 sword) that no fixed-persona baseline solves jointly, with value loss held at ≈0.04 — matching the most stable specialist. Accepted as a work-in-progress paper; full multi-seed validation planned.

PEAK – Platformer Engine for RL Benchmarking | *Python, Gymnasium, Stable-Baselines3, Hydra* — 2025–2026
- Co-developed a deterministic high-performance DRL benchmarking engine with modular persona-based reward shaping, evaluating PPO agents across 11 custom 2D platformer stages with reproducible SMB1-style physics. GitHub: Code-SorceryLab/PEAK-DRL-Tool.
- Built a **Dual Spatial Hashing** collision system (separate static/dynamic grids) achieving O(C) per-query cost and sustained **1000+ environment steps per second**, making 10M-step training runs practical on single-machine hardware.

Gestura – Gesture-Based Medical Screening | *Python, Flask, CV, Gemini API, Three.js* — Jan 2026
- Built at **HackHive 2026** a non-verbal patient intake tool: Flask backend streaming real-time hand-tracking state to a Three.js 3D "digital twin" frontend, with **Google Gemini** converting gestures into SOAP-style clinician summaries.

Technical Skills

Languages: Python, C++, SQL, Bash, JavaScript (Node.js/Three.js), HTML/CSS, P4
ML / RL: PyTorch, Stable-Baselines3, Gymnasium, PPO, Crafter, TensorBoard, Hydra, NumPy, Pandas, Scikit-Learn
AI Systems & Web: Gemini API, Ollama, LLM prompt engineering, Computer Vision, Flask, REST APIs, Docker, Git
Networking & Sec: Intel Tofino P4, Wireshark, Nmap, Cisco IOS, Linux (Kali/RedHat), Vulnerability Assessment

Certifications: Cisco Certified Network Associate (CCNA) 2024–2027 • TryHackMe 200+ Hours (2023–Present) • Dean's List 2024-2026